End Semester Examination 2024

Name of the Program:Diploma | Semester: 6th

Name of theSubj :Computer Graphics | CourseCode:DTCS-606

Time:3Hour | Maximum Marks:100

Note:

(i) Answer **all the questions** by choosing ***any two of the subquestions.***
(ii) Each question carries 10marks.
(iii) Total marks in each main question are twenty.
(iv) All questions are compulsory.

Q1	**(20 marks)**	
(a)	Write about following: a. Image b. Pixels c. Resolution d. Aspect Ration	CO1
(b)	Explain the latest trends in virtual reality and augmented reality. Explain the applications of both.	
(c)	Expalin the following: a. Raster scan display b. Random scan display c. Flat panel display	
Q2	**(20 marks)**	
(a)	Write the differences between Flood fill and Boundary fill algorithms	CO2
(b)	Explain the character generation methods – Stroke method, bitmap ,starbust method.	
(c)	Write the DDA algorithm for Line Generation.	
Q3	**(20 marks)**	
(a)	Explain the parallel and perspective projection. Draw a chart for classification of projections.	CO3
(b)	Explain the 2D Scaling and Translation transformation.	
(c)	Provide the homogeneous representation for following transformations in 2d a. Translation b. Scaling c. Rotation d. Shearing	
Q4	**(20 marks)**	
		CO4

(a)	Explain the following terms: a. Windowing b. Clipping c. Viewport	
(b)	Explain the Cohen Sutherland Line Clipping algorithm	
(c)	Explain the different methods of text clipping	
	(20 marks)	
Q5		
(a)	What is Bezier curve? Write the properties of Bezier curve?	CO5
(b)	Explain different types of representation of curves –Implicit , Explicit, and parameteric curves.	
(c)	Explain the DDA algorithm for arc generation	